



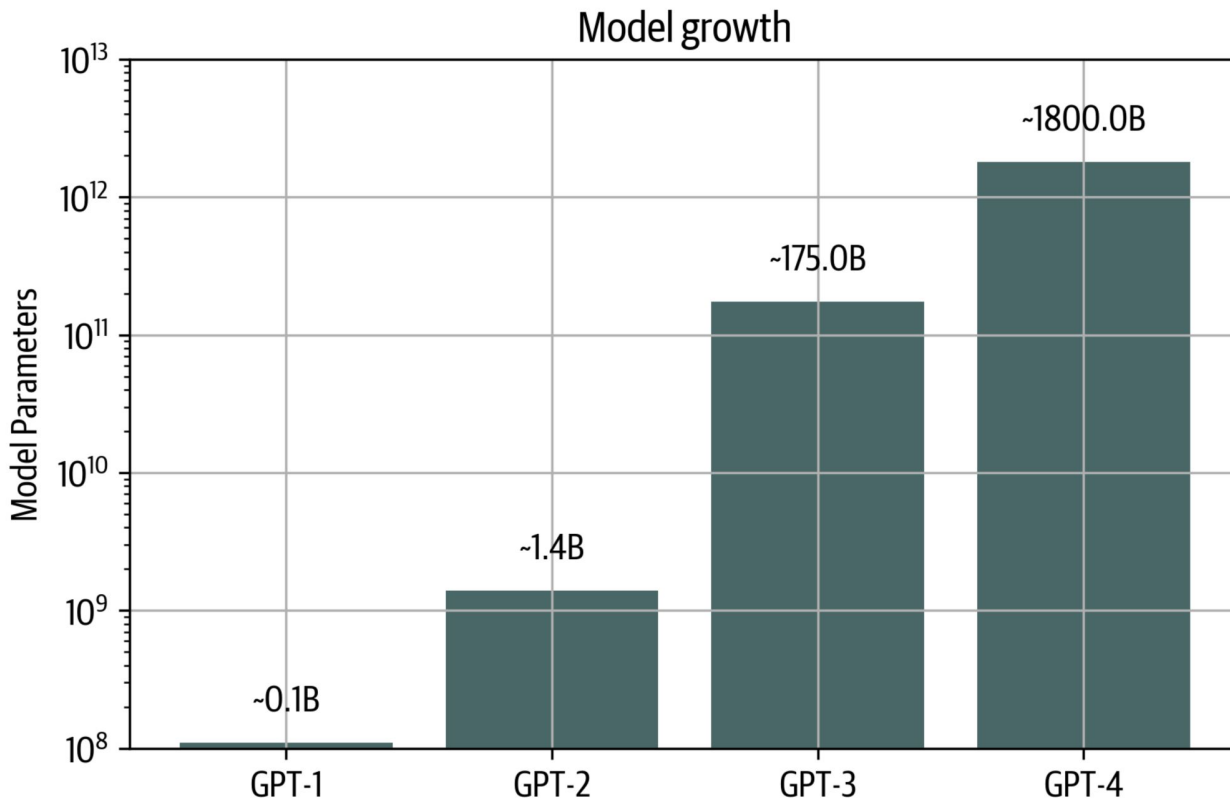
Training Data for LLMs

Leandro von Werra

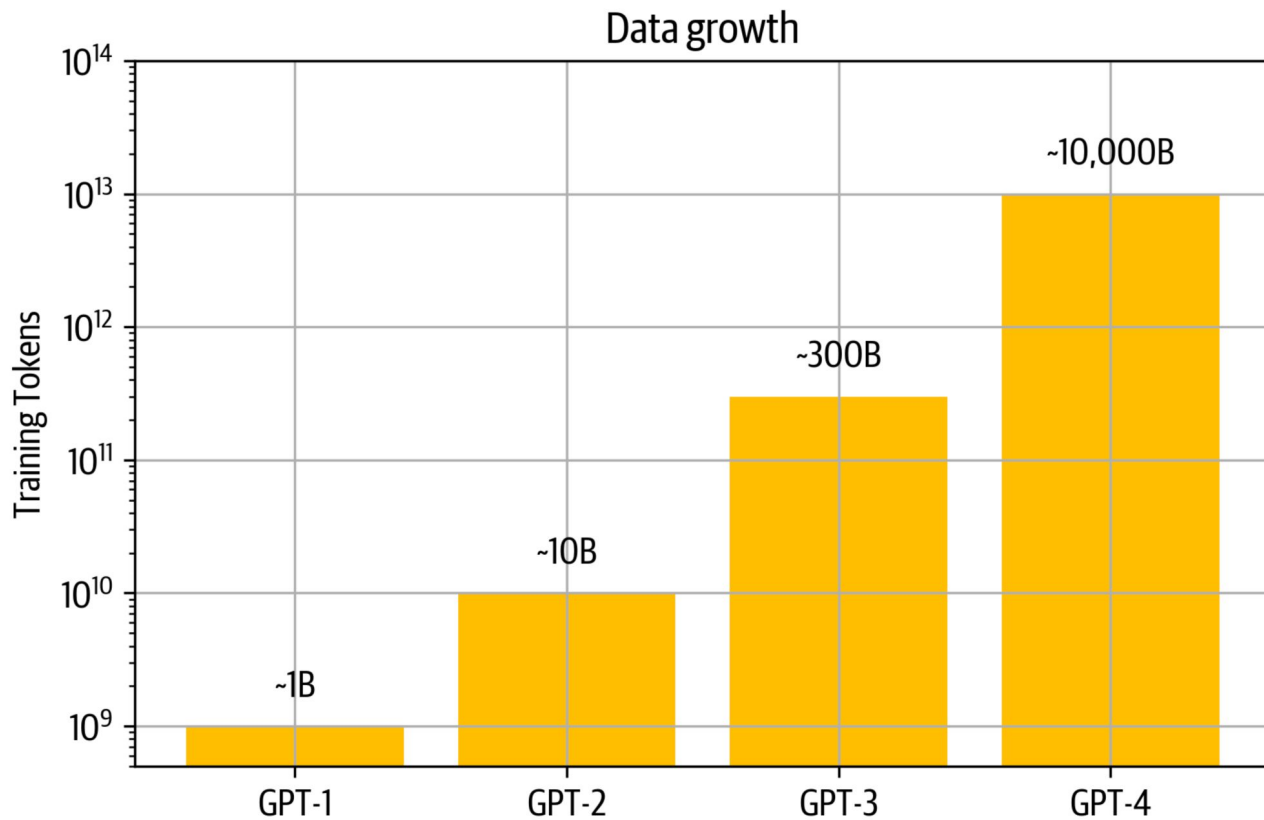
State of LLMs

The scaling era

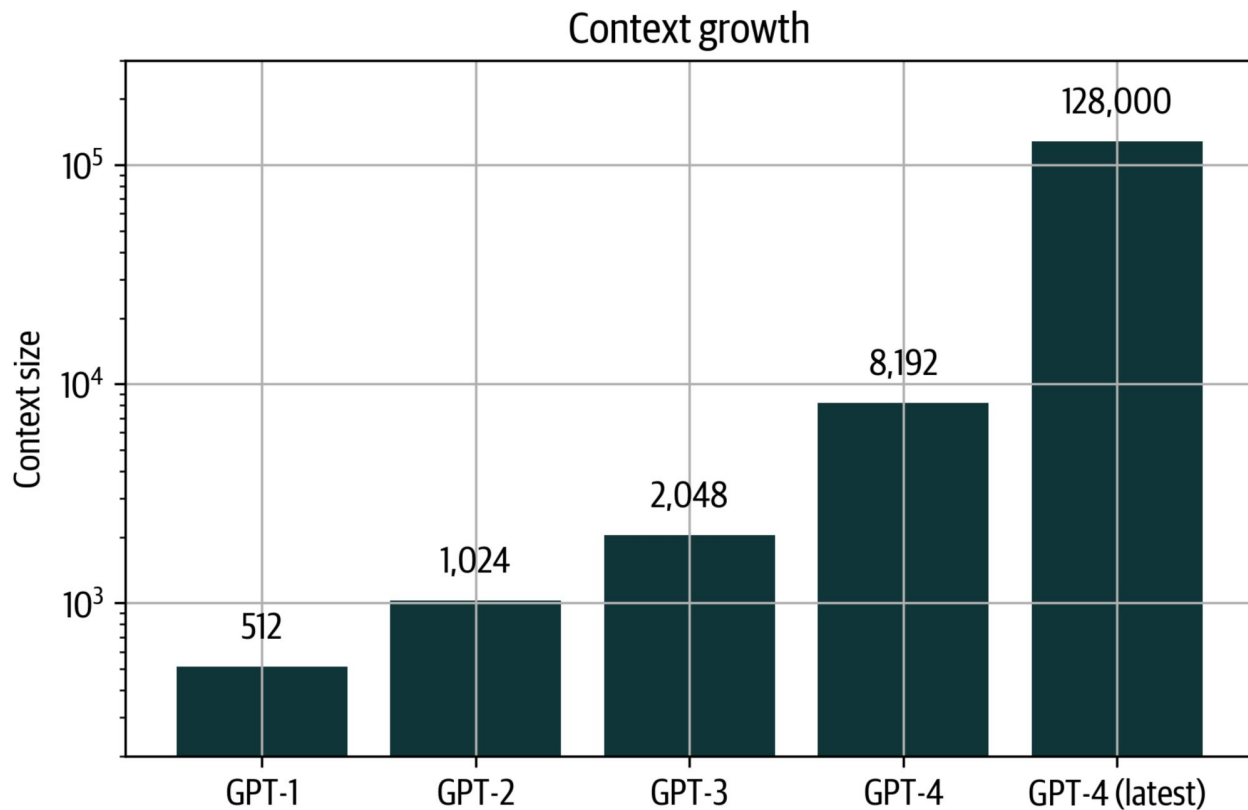
Trends: **train larger**



Trends: train longer



Trends: **more context**



Datasets

The secret sauce
and 90% of all the work

Dataset: the secret workhorse of LLMs

The “it” in AI models is the dataset.

Posted on June 10, 2023 by jbetker

I've been at OpenAI for almost a year now. In that time, I've trained a lot of generative models. More than anyone really has any right to train. As I've spent these hours observing the effects of tweaking various model configurations and hyperparameters, one thing that has struck me is the similarities in between all the training runs.

It's becoming awfully clear to me that these models are truly approximating their datasets to an incredible degree. What that means is not only that they learn what it means to be a dog or a cat, but the interstitial frequencies between distributions that don't matter, like what photos humans are likely to take or words humans commonly write down.

What this manifests as is – trained on the same dataset for long enough, pretty much every model with enough weights and training time converges to the same point. Sufficiently large diffusion conv-unets produce the same images as VIT generators. AR sampling produces the same images as diffusion.

This is a surprising observation! It implies that **model behavior is not determined by architecture, hyperparameters, or optimizer choices. It's determined by your dataset, nothing else.** Everything else is a means to an end in efficiently delivery compute to approximating that dataset.

Then, when you refer to “Lambda”, “ChatGPT”, “Bard”, or “Claude” then, it's not the model weights that you are referring to. It's the dataset.

<https://nonint.com/2023/06/10/the-it-in-ai-models-is-the-dataset/>



Amanda Askill @AmandaAskill · Mar 24

"Is this behavior emergent or does it come from the data?" is not a debate we should be having. All emergent behavior comes from the data. It's true of humans and it's true of AI. None of us has ever magically pulled anything out of the ether.

29

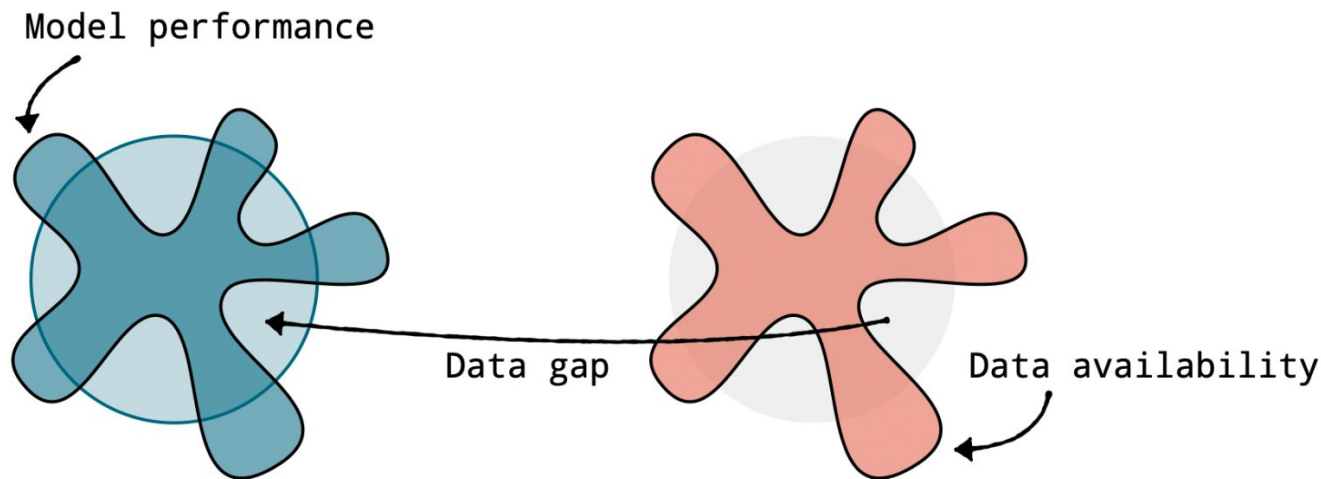
35

247

56K



Dataset: jagged AI frontier is a data frontier



Dataset: **goal of pretraining**

Train a general-purpose model → maximal coverage

Requires:

- train on massive quantities of text, at least 1 trillion of tokens nowadays moving towards 10-20T

Challenges:

- maximizing diversity and coverage
- maximising quality, robustness
- data quality evaluation: how to measure data quality at the billion tokens scale

Dataset: where to find data

- Very large ($> 100\text{B}$ tokens):
 - **Common crawl**: everyone starts from here
 - **Code**: Github and Software Heritage
- Curated:
 - **Websites**: Wikipedia, StackExchange, Arxiv
 - **Books**: public-domain vs. copyrighted
- More recent trends
 - **Synthetic data**

Dataset: **the average web**

... is mostly garbage:

- Ads/SEO
- Obituaries
- Porn
- Sport News

If we want a high quality model we need to clean it up!

Dataset: **general strategy**

- Gather as much data as possible
- Filter as much as necessary
- Look at the data you keep (and throw away)
(manually, clustering, tokenizing etc)
- Don't trust your intuition → evaluate!

Dataset: filtering pipeline

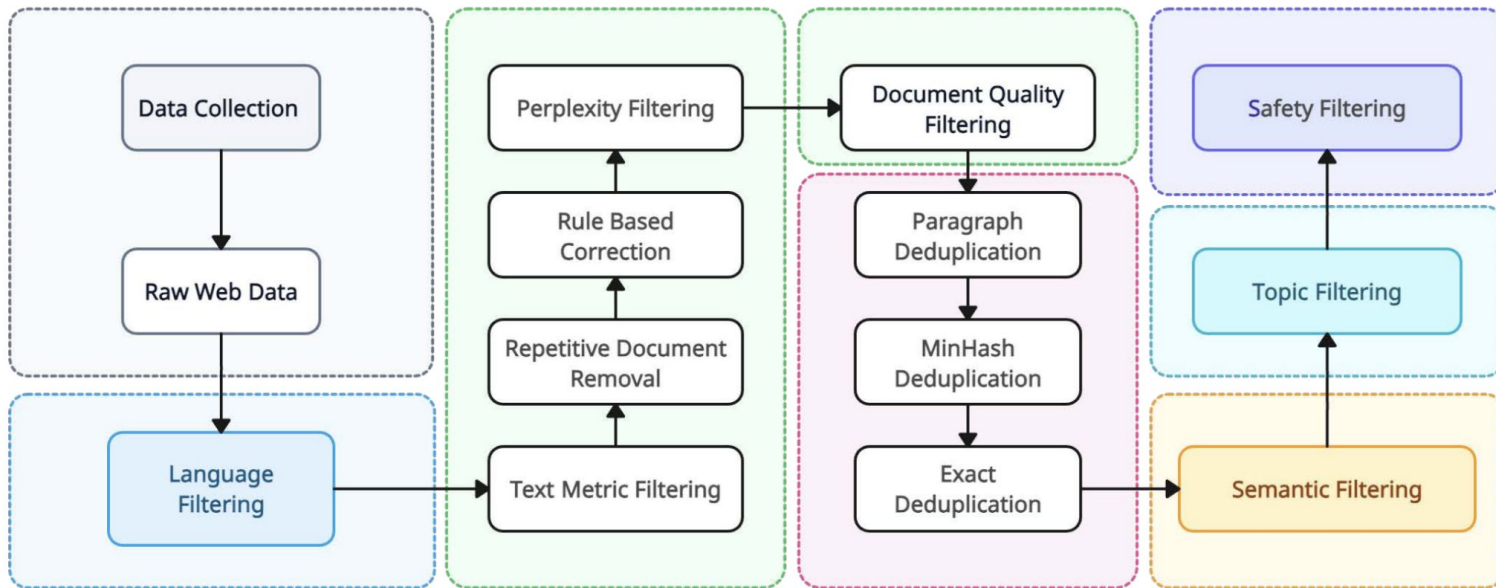


Figure 1: Yi’s pretraining data cleaning pipeline.

Dataset: filtering pipeline

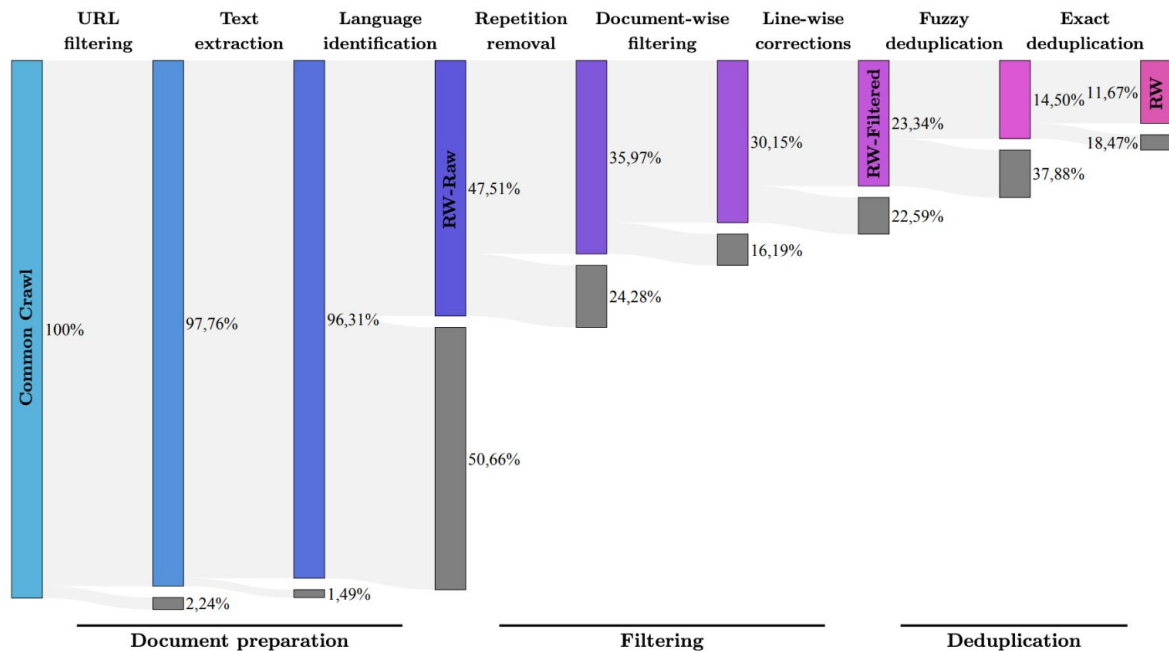


Figure 2. Subsequent stages of Macrodata Refinement remove nearly 90% of the documents originally in CommonCrawl. Notably, filtering and deduplication each result in a halving of the data available: around 50% of documents are discarded for not being English, 24% of remaining for being of insufficient quality, and 12% for being duplicates. We report removal rate (grey) with respect to each previous stage, and kept rate (shade) overall. Rates measured in % of documents in the document preparation phase, then in tokens.

Dataset: language filtering

*fast*Text

Language Filtering Summary & Ideas

- Language filtering is an important first step of data selection where documents are selected to include only the desired languages.
 - Classifier-based methods are the norm when creating both english-only and multilingual datasets.
 - For lower-resourced languages, URL-based methods are also useful for classifying the language used in multilingual datasets and can sometimes can be more reliable than classifiers.
 - Filtering for code languages is very simple, most works simply using the file extension.
- 💡 One interesting direction for future research is developing methods that find code data that within documents of natural language. To the best of our knowledge, no such methods currently exist, but a code-language identifier could be used in domains with interleaved code and natural language to find meaningful quantities of new code data.



 Hugging Face  Model  Hugging Face  Space (Demo)  Hugging Face  Data  license  Apache-2.0  Stars  88  arXiv  2310.16248

Language Identification with Support for More Than 2000 Labels

TL;DR: The repository introduces **GlottLID**, an open-source language identification (LID) model with support for more than **2000 labels**.

Latest: GlottLID is now updated to V3. V3 supports 2102 labels (three-letter [ISO 639-3](#) codes with script). For more details on the supported languages and performance, as well as significant changes from previous versions, please refer to [languages-v3.md](#).

Dataset: quality heuristics

Heuristic Category	Common Utility Functions	Example Selection Mechanisms
Item Count	# of characters in a {word/line/paragraph/document} # of {words/lines/sentences} in a document	Remove documents with fewer than 5 words (Raffel et al., 2020)
Repetition Count	# of times a {character/n-gram/word/sentence/paragraph} is repeated	Remove lines that repeat the same word more than 4 times consecutively (Laurençon et al., 2022)
Existence	Whether a {word/n-gram} is in the document Whether a terminal punctuation is at the end of a line	Remove lines starting with “sign-in” (Penedo et al., 2023)
Ratio	% of alphabetic characters in a document % of numerals/uppercase characters in a {line/document}	Remove documents with a symbol-to-word ratio greater than 0.1 (for “#” and “...”) (Rae et al., 2022)
Statistics	The mean length (and standard deviation) of all lines in a document	Remove code files that have mean line length greater than 100 characters (Chen et al., 2021)

Table 2: Commonly used heuristic utility functions and demonstrative selection mechanisms.

Dataset: **quality** heuristics

Advantages:

- controlled
- robust
- deterministic
- rather clear priors

Drawbacks:

- rely entirely on surface level
- danger of removing too much
- hyper-parameter tuning

Dataset: **quality filtering** - ML

Given a set of examples of good/bad documents:

- **classifier-based quality filtering:**
fastText classification with an n-gram size of 2
- **perplexity based filtering:**
5-gram Kneser-Ney model on Wikipedia
(see <https://github.com/kpu/kenlm>)

→ Filter based on a threshold

- more “quality/content based filtering”
- harder to estimate the impact → may introduce “bias”

Dataset: notes on filtering

Taking care of domains specificities

- important to inspect the effect on domain specific data
- extract 10 documents per domains (e.g. top urls)
- manually inspect the results
- craft domain specific filters/hyper-parameters
- same for multiple languages

Deterministic vs. stochastic selection

- hard threshold are strong decision points
- stochastic smoothing of rules

Dataset: deduplication

Fuzzy

- BLOOM filters (hashing and fixed size vector)
- MinHash (hashing and sorting)

Exact

- Exact substring with suffix array
- Sentence dedup

time/memory consumption

- MinHash offers a good trade-off of speed/memory

counter intuitive results

- more deduplication may lead to keeping only bad data

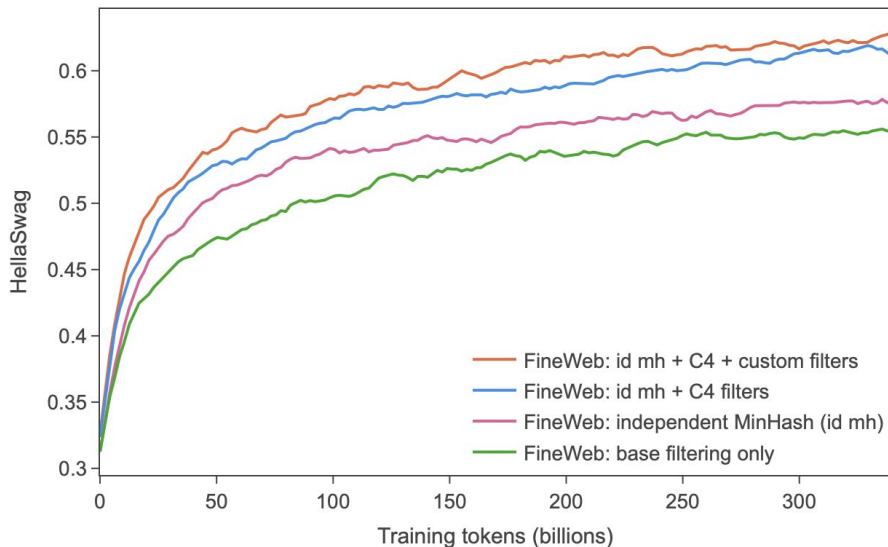
Dataset: **evaluate data quality**

Small models trainings: **train 1-2B size models on 30B tokens** (chinchilla optimal)

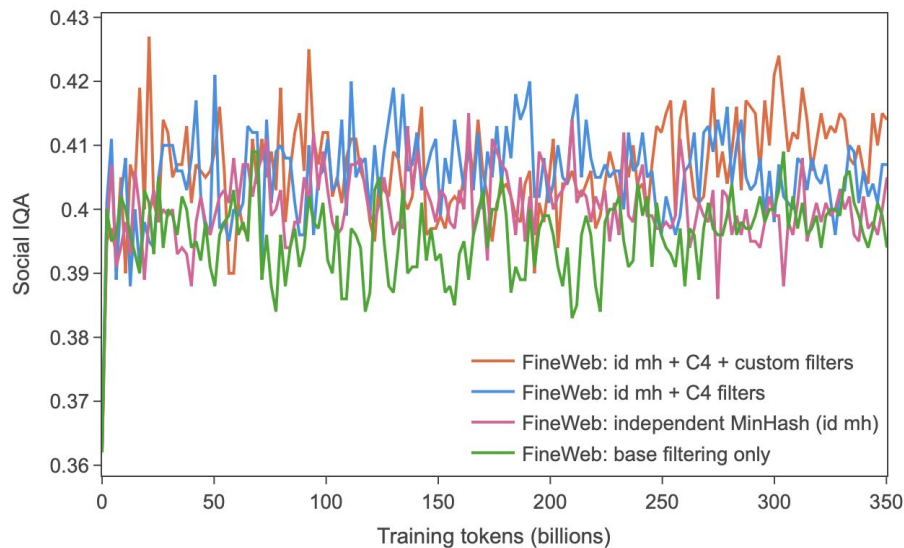
- Use a set of “high-signal” benchmarks (in NLP):
 - commonsense QA
 - hellaswag
 - openbook QA
 - PiQA
- High-signal?
 - **monotonicity**: monotonically increasing during training
 - **low variance**:
 - when comparing two known reference datasets (e.g. The Pile versus C4)
 - when comparing with various sub-parts of data and seeds
 - above random baseline
- Tricky details to maximize signal:
 - Small models like “normalized loglikelihood” better
 - Larger models like “letter answers” better

Dataset: evaluate data quality

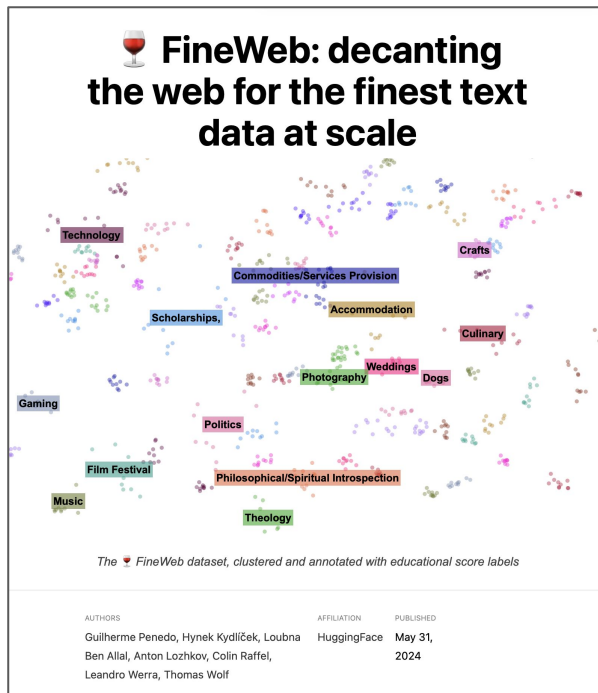
The different FineWeb processing steps



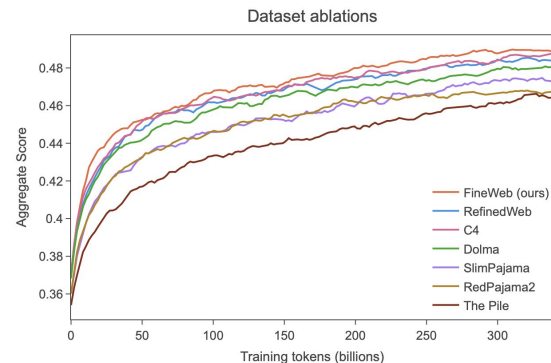
The different FineWeb processing steps



Dataset: FineWeb



- Based on CommonCrawl
- 44TB disk space
- 15T tokens
- Transparent pipeline



<https://hf.co/spaces/HuggingFaceFW/blogpost-fineweb-v1>

Dataset: FineWeb-Edu

Below is an extract from a web page. Evaluate whether the page has a high educational value and could be useful in an educational setting for teaching from primary school to grade school levels using the additive 5-point scoring system described below. Points are accumulated based on the satisfaction of each criterion:

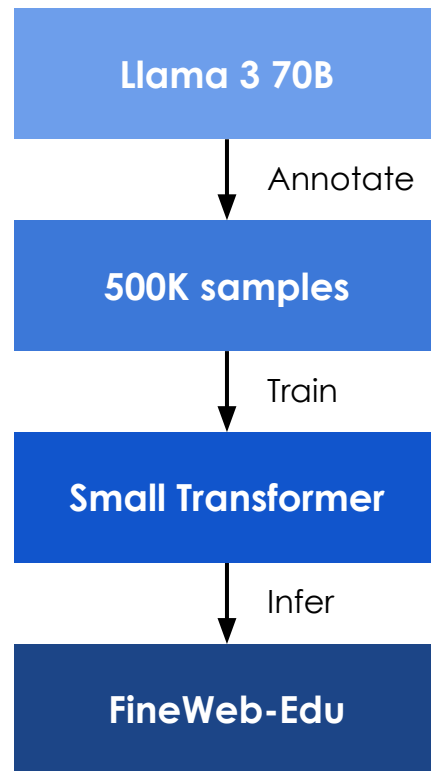
- Add 1 point if the extract provides some basic information relevant to educational topics, even if it includes some irrelevant or non-academic content like advertisements and promotional material.
- Add another point if the extract addresses certain elements pertinent to education but does not align closely with educational standards. It might mix educational content with non-educational material, offering a superficial overview of potentially useful topics, or presenting information in a disorganized manner and incoherent writing style.
- Award a third point if the extract is appropriate for educational use and introduces key concepts relevant to school curricula. It is coherent though it may not be comprehensive or could include some extraneous information. It may resemble an introductory section of a textbook or a basic tutorial that is suitable for learning but has notable limitations like treating concepts that are too complex for grade school students.
- Grant a fourth point if the extract is highly relevant and beneficial for educational purposes for a level not higher than grade school, exhibiting a clear and consistent writing style. It could be similar to a chapter from a textbook or a tutorial, offering substantial educational content, including exercises and solutions, with minimal irrelevant information, and the concepts aren't too advanced for grade school students. The content is coherent, focused, and valuable for structured learning.
- Bestow a fifth point if the extract is outstanding in its educational value, perfectly suited for teaching either at primary school or grade school. It follows detailed reasoning, the writing style is easy to follow and offers profound and thorough insights into the subject matter, devoid of any non-educational or complex content.

The extract:

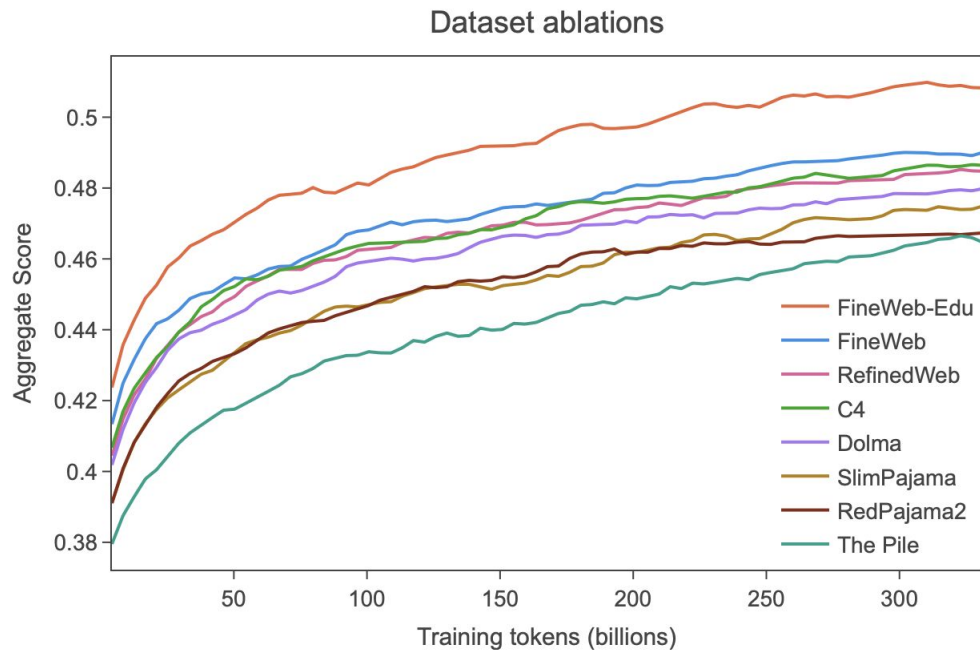
<EXAMPLE>.

After examining then extract:

- Briefly justify your total score, up to 100 words.
- Conclude with the score using the format: "Educational score: <total points>"



Dataset: FineWeb-Edu

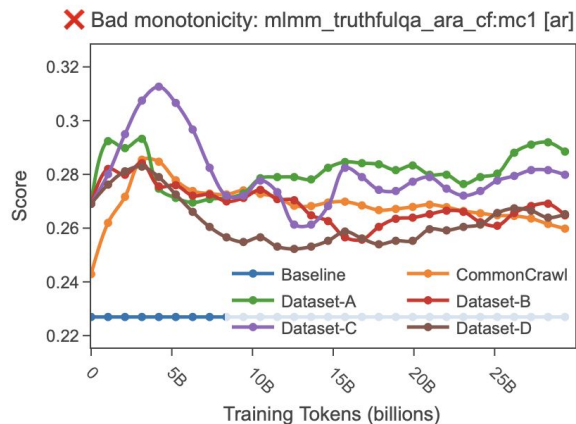
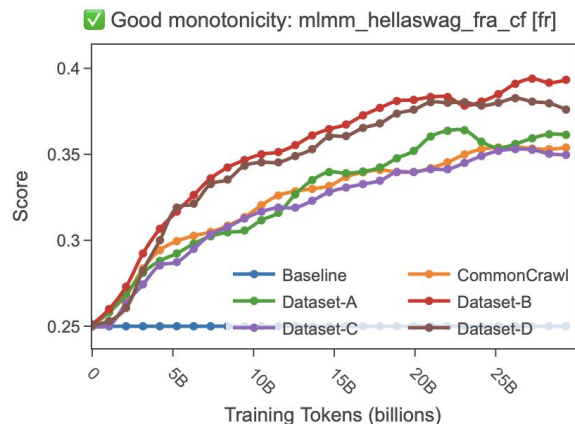
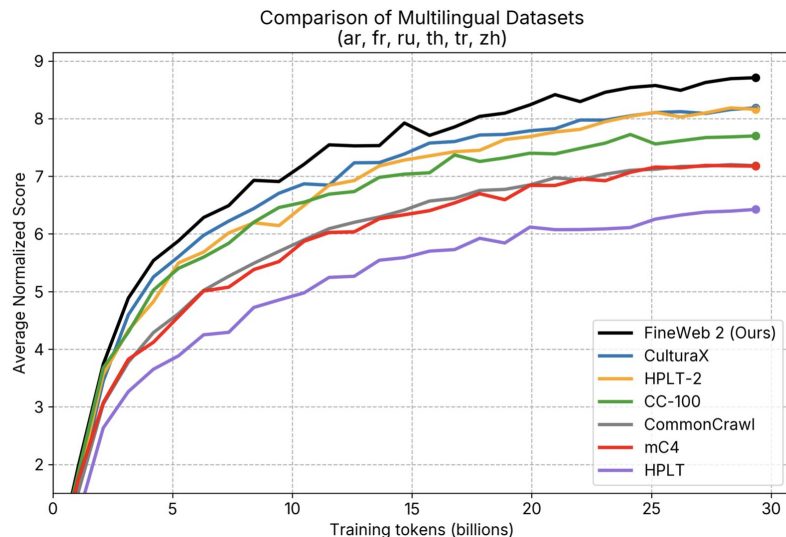


Dataset: where can we get more?

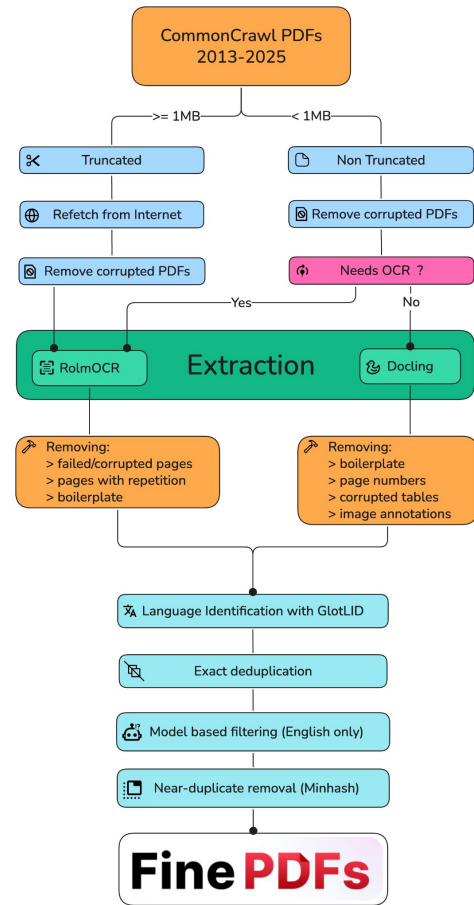
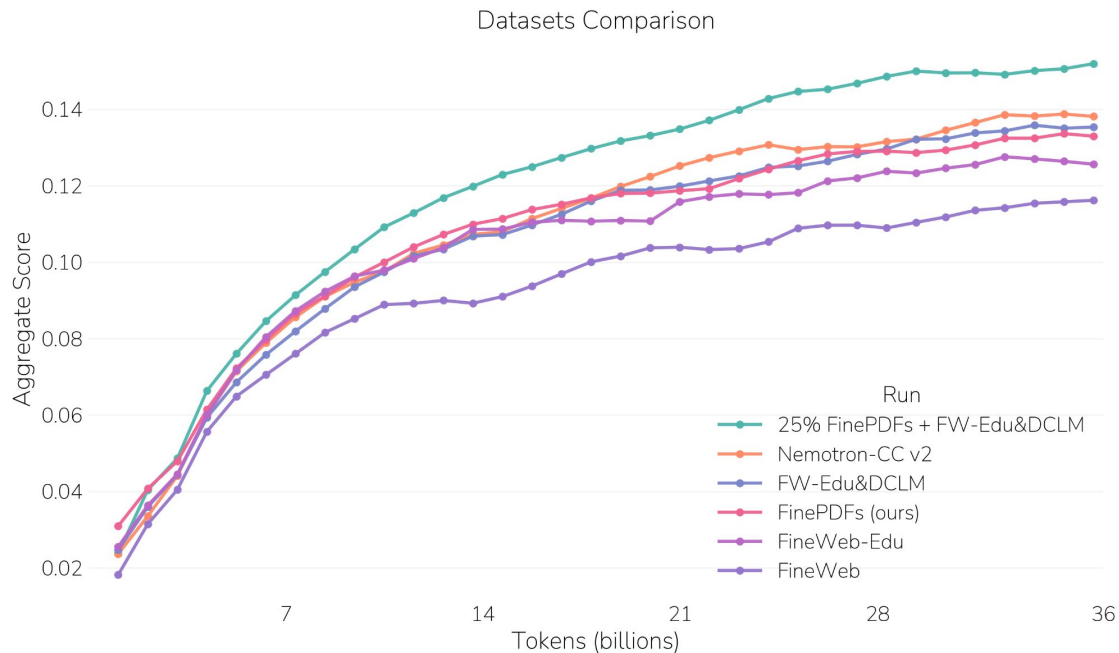
Dataset: FineWeb2

FineWeb2: One Pipeline to Scale Them All — Adapting Pre-Training Data Processing to Every Language

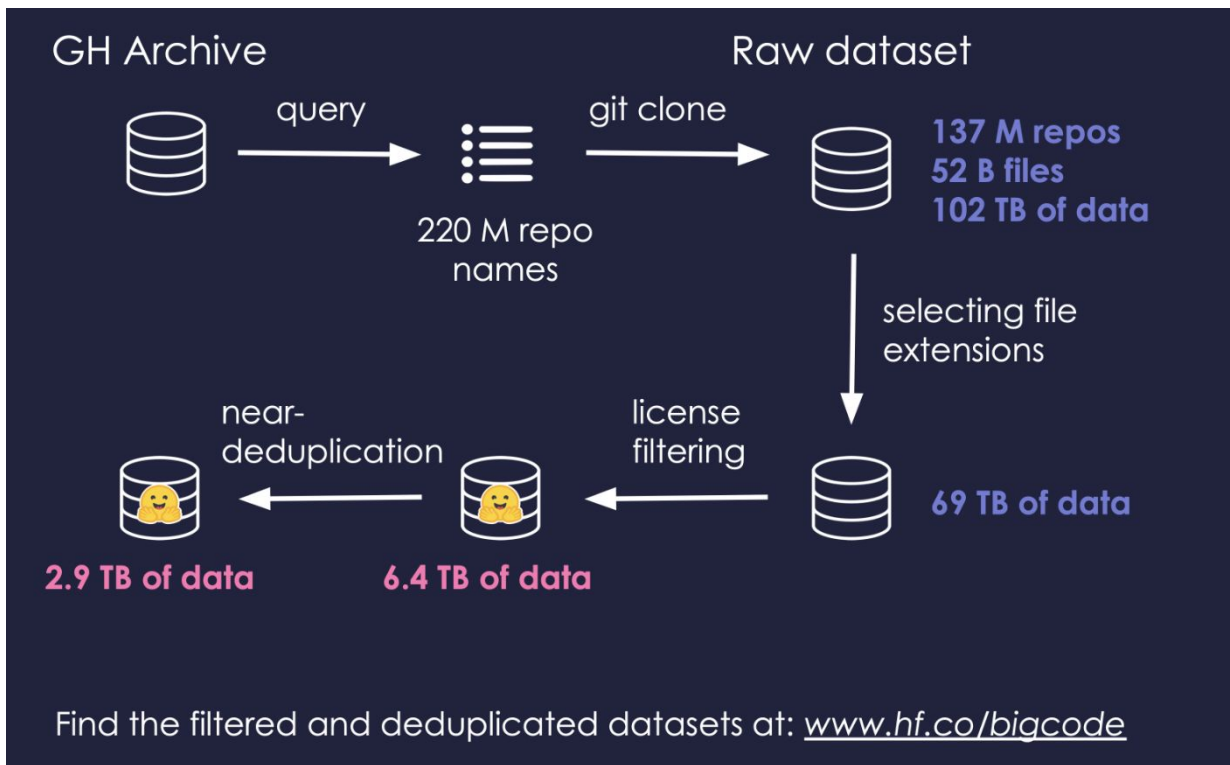
Guilherme Penedo^{EPFL}* Hynek Kydlíček^{EPFL} Vinko Sabolčec^{EPFL} Bettina Messmer^{EPFL}
Negar Foroutan^{EPFL} Amir Hossein Kargaran^{EPFL} Colin Raffel^{EPFL} Martin Jaggi^{EPFL}
Leandro Von Werra^{EPFL} Thomas Wolf^{EPFL}



Dataset: FinePDFs



Dataset: The Stack



Dataset: The Stack v2

- 3B+ files in 658 programming languages
- created as part of the BigCode Project pre-training dataset for Code LLMs
- Derived from the Software Heritage archive: largest public archive of software source code

	The Stack v1	The Stack v2
full	6.4TB	67.5TB
dedup	2.9TB	32.1TB
train (full)	~200B tokens	~900B tokens

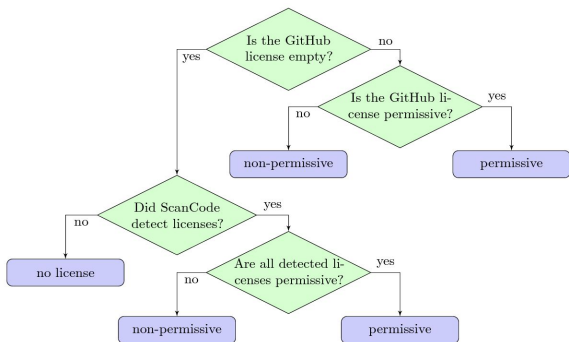


Figure 1: File-level license assignment logic.

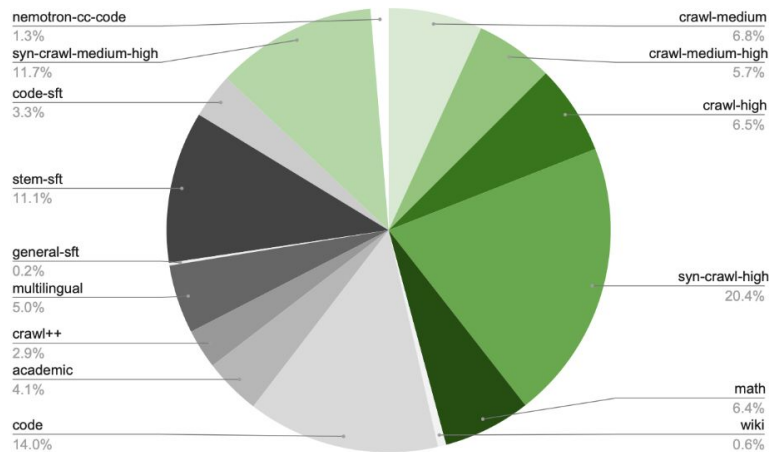
StarCoder 2 and The Stack v2: The Next Generation

Published on Feb 29 · ★ Featured in [Daily Papers](#) on Mar 1

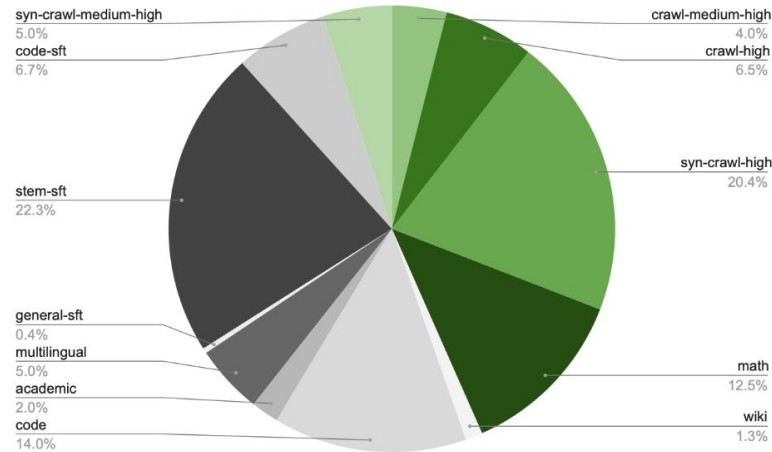
Authors: Anton Lozhkov, Raymond Li, Loubna Ben Allal, Federico Cassano, Joel Lamy-Poirier, Nouamane Tazi, Ao Tang, Dmytro Pykhtar, Jiawei Liu, Yuxiang Wei, Tianyang Liu, Max Tian, Denis Kocetkov, Arthur Zucker, Younes Belkada, Zijian Wang, Qian Liu, Dmitry Abulkhanov, Indraneil Paul, Zhuang Li, Wen-Ding Li, Megan Risdal, Jia Li, Jian Zhu, Terry Yue Zhuo, Evgenii Zheltonozhskii, Nii Osae Osae Dade, Wenhao Yu, Lucas Krauß, Naman Jain, Yixuan Su, Xuanli He, Manan Dey, Edoardo Abati, Yekun Chai, Niklas Muennighoff, Xiangru Tang, Muhtasham Oblokulov, Christopher Akiki, Marc Marone, Chenghao Mou, Mayank Mishra, Alex Gu, Binyuan Hui, Tri Dao, Armel Zebaze, Olivier Dehaene, Nicolas Patry, Canwen Xu, Julian McAuley, Han Hu, Torsten Scholak, Sebastien Paquet, Jennifer Robinson, Carolyn Jane Anderson, Nicolas Chapados, Mostofa Patwary, Nima Tajbakhsh, Yacine Jernite, Carlos Muñoz Ferrandis, Lingming Zhang, Sean Hughes, Thomas Wolf, Arjun Guha, Leandro von Werra, Harm de Vries

Mid/Post-training: Nemotron 3

Nemotron 3 Nano : Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning



(a) Data mixture of Phase 1.



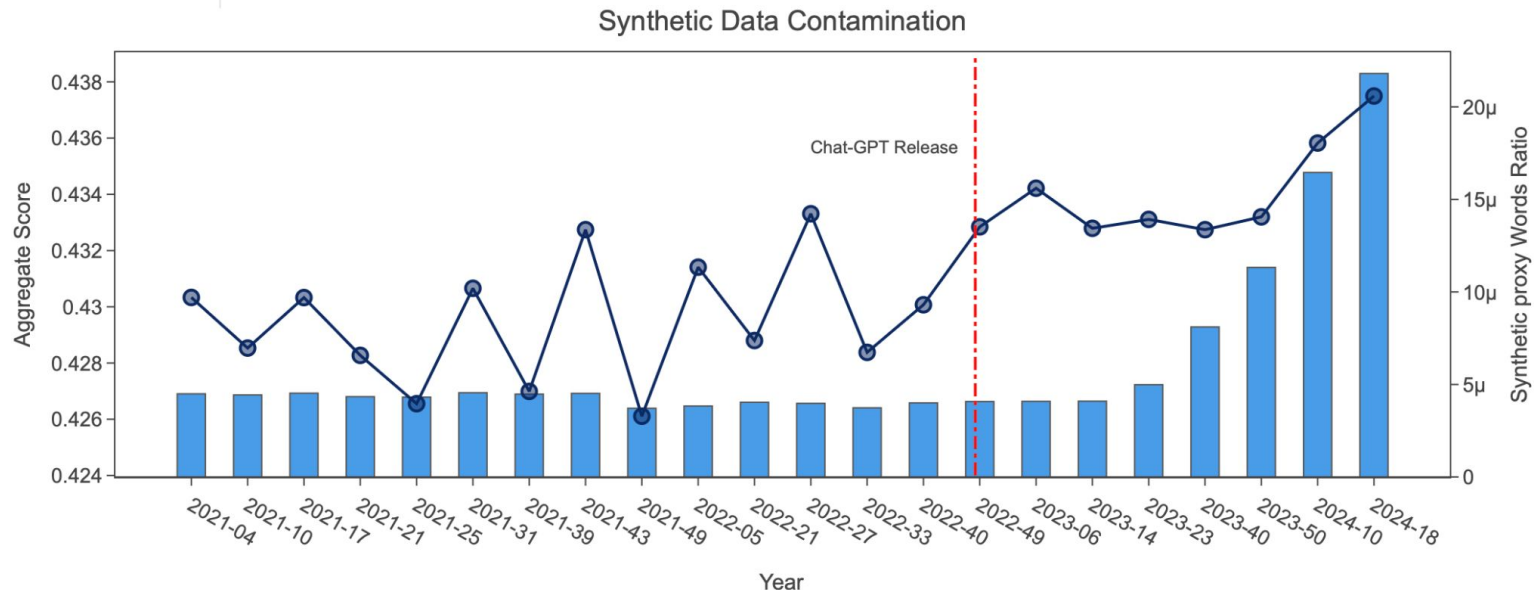
(b) Data mixture of Phase 2.

diversity



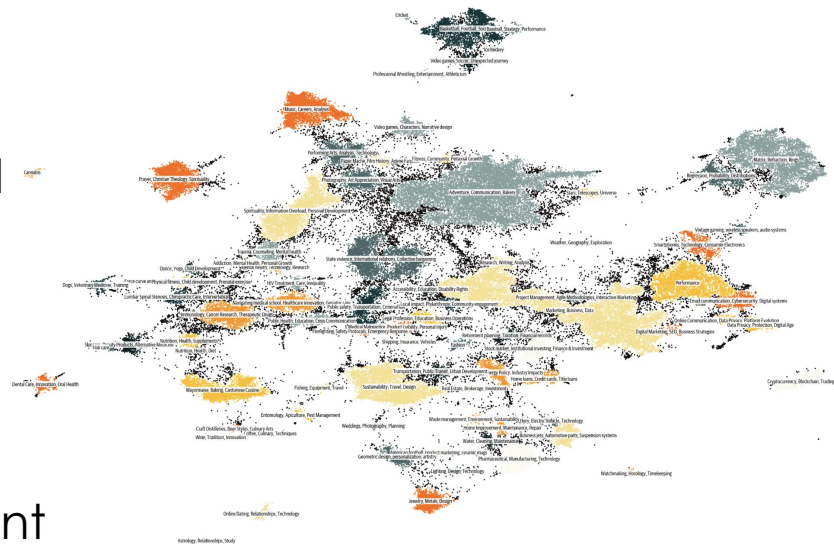
quality

Dataset: Synthetic Data?



Dataset: Cosmopedia

- A synthetic dataset of 30M samples
- generated by Mixtral-8x7B-Instruct-v0.1
- 8 splits:
 - various sources for seed samples: Stanford, OpenStax and KhanAcademy, FineWeb, instruction-tuning datasets
 - model is asked to generate content



Dataset: FinePhrases

FINEWEB-EDU

916 words · 5,501 chars

A Conversation with Physicist Y.C. Chen
A technological breakthrough in imaging that has applications in early diagnosis of eye ailments was the fruit of a recent collaboration between Professor Y.C. Chen, a laser specialist who is chairman of the Hunter Physics Department, and Dr. Ronald H. Silverman, then of the Weill Cornell Medical College Ophthalmology Department.
Silverman, a specialist in ultrasound imaging of the eye who is now at Columbia, was exploring the possibilities of combining ultrasound with light to improve image resolution; Chen wanted to apply his laser knowledge to the biomedical field. Together, the two invented a device that creates a detailed image of the eye, perhaps 20 times better than the images available with ultrasound alone - allowing doctors to see early signs of macular degeneration, tumors, oxygen shortages, and other danger signals.

ID: <urn:uuid:311230c5-de62-4cf8-960d-a8345c8cdb51>

Original Website: <http://www.hunter.cuny.edu/communications/news/top-featured-stories/a-conversation-with-physicist-y.c.-chen>

CommonCrawl S3: s3://commoncrawl/crawl-data/CC-MAIN-2013-20/segments/1368696383156/warc/CC-MAIN-20130516092623-00002-ip-10-60-113-184.ec2.internal.warc.gz

FAQ

182 words · 1,149 chars

A technological breakthrough in imaging with applications in early diagnosis of eye ailments was the result of a recent collaboration between Professor Y.C. Chen, a laser specialist, and Dr. Ronald H. Silverman, who was a specialist in ultrasound imaging of the eye at the time. Chen's interest in applying laser technology to the biomedical field led them to develop a device that creates a detailed image of the eye 20 times better than the images available with ultrasound alone. This device allows doctors to detect early signs of macular degeneration, tumors, oxygen shortages,

TABLE

152 words · 1,065 chars

Table:

Year	Research Project
2008	Development of Photoacoust
2008	Development of Photoacoust
2016	Development of Non-transpa

Question-Answer Pair:

How does the photoacoustic imaging process work?

MATH

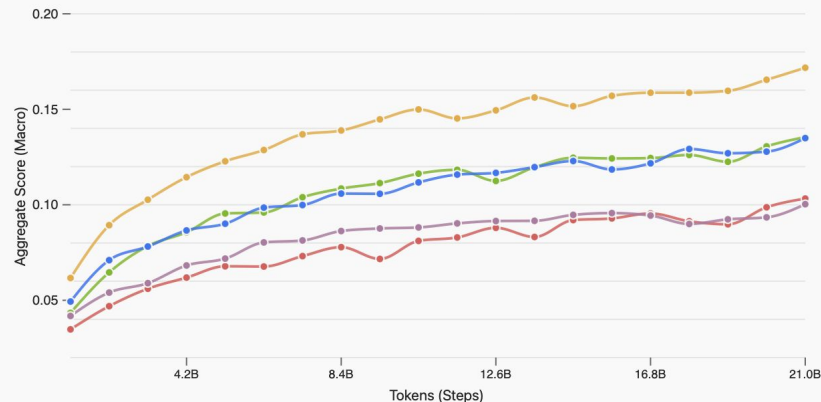
45 words · 310 chars

What is the photoacoustic imaging process, and how does it improve resolution about 20 times? Answer: The photoacoustic imaging process uses ultrasound generated by light, which is reflected and processed by a computer to create high-resolution images of tissue, revealing microvasculature and the optic nerve.

TUTORIAL

44 words · 251 chars

To use a laser to generate an ultrasound image, first, focus a laser beam into a point. When a tissue absorbs the laser, it warms up, expands, and generates an ultrasound. This improves the resolution by at least 20 times compared to ultrasound alone.



VIEW

Training Progression

METRIC

Aggregate Score (Macro)

Legend

FinePhrase (table) Nemotron-HQ-Synth REWIRE Cosmopedia SYNTH

Questions

GitHub/HF Hub/X: [lvwerra](#)